

Exponential Family Distributions

Exponential Family - Standard Form

$$f(y|\theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

Key Properties:

- θ : natural (canonical) parameter
- ϕ : dispersion parameter
- $b(\theta)$: cumulant function (log-partition function)
- **Mean:** $E(Y) = \mu = b'(\theta)$
- **Variance:** $\text{Var}(Y) = \phi b''(\theta) = \phi V(\mu)$
- **Variance Function:** $V(\mu) = b''((b')^{-1}(\mu))$
- **MGF:** $M_Y(t) = \exp \left\{ \frac{b(\theta + \phi t) - b(\theta)}{\phi} \right\}$
- **Natural link:** $g = (b')^{-1}$, gives $\theta = g(\mu)$

Normal Distribution $N(\mu, \sigma^2)$

- PDF: $f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y-\mu)^2}{2\sigma^2} \right\}$
- Support: $y \in (-\infty, \infty)$
- Mean: $E(Y) = \mu$, Variance: $\text{Var}(Y) = \sigma^2$
- MGF: $M(t) = \exp \left(\mu t + \frac{\sigma^2 t^2}{2} \right)$

Exp Form: $f(y|\mu, \sigma^2) = \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right\}$

- $\theta = \mu$, $b(\theta) = \theta^2/2$, $\phi = \sigma^2$, $V(\mu) = 1$, Link = Identity
- Log-lik: $\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu)^2$
- Score: $U(\mu) = \frac{n(\bar{y} - \mu)}{\sigma^2}$, Fisher: $\mathcal{I}(\mu) = \frac{n}{\sigma^2}$

Bernoulli Distribution $Ber(p)$

- PMF: $f(y) = p^y(1-p)^{1-y}$, $y \in \{0, 1\}$
- Mean: $E(Y) = p$, Variance: $\text{Var}(Y) = p(1-p)$
- MGF: $M(t) = 1 - p + pe^t$

Exp Form: $f(y|p) = \exp \left\{ y \log \frac{p}{1-p} + \log(1-p) \right\}$

- $\theta = \log \frac{p}{1-p}$ (log-odds), $p = \frac{e^\theta}{1+e^\theta}$
- $b(\theta) = \log(1 + e^\theta)$, $\phi = 1$, $V(\mu) = \mu(1-\mu)$, Link = Logit
- Log-lik: $\ell(p) = \sum [y_i \log p + (1 - y_i) \log(1 - p)]$
- Score: $U(p) = \frac{\sum y_i - np}{p(1-p)}$, Fisher: $\mathcal{I}(p) = \frac{n}{p(1-p)}$

Binomial Distribution $Bin(k, p)$

- PMF: $f(y) = \binom{k}{y} p^y (1-p)^{k-y}$, $y \in \{0, 1, \dots, k\}$
- Mean: $E(Y) = kp$, Variance: $\text{Var}(Y) = kp(1-p)$
- MGF: $M(t) = (1 - p + pe^t)^k$

Exp Form: $f(y|p) = \exp \left\{ y \log \frac{p}{1-p} + k \log(1-p) + \log \binom{k}{y} \right\}$

- $\theta = \log \frac{p}{1-p}$, $b(\theta) = k \log(1 + e^\theta)$, $\phi = 1$
- $V(\mu) = \mu(1 - \mu/k)$, Link = Logit
- For proportion $\hat{p} = Y/k$: $E(\hat{p}) = p$, $\text{Var}(\hat{p}) = \frac{p(1-p)}{k}$
- Log-lik: $\ell(p) = \sum y_i \log p + (nk - \sum y_i) \log(1-p) + \text{const}$
- Score: $U(p) = \frac{\sum y_i - nkp}{p(1-p)}$, Fisher: $\mathcal{I}(p) = \frac{nk}{p(1-p)}$

Poisson Distribution $Pois(\lambda)$

- PMF: $f(y) = \frac{\lambda^y e^{-\lambda}}{y!}$, $y \in \{0, 1, 2, \dots\}$
- Mean: $E(Y) = \lambda$, Variance: $\text{Var}(Y) = \lambda$ (mean = var!)
- MGF: $M(t) = \exp\{\lambda(e^t - 1)\}$

Exp Form: $f(y|\lambda) = \exp\{y \log \lambda - \lambda - \log(y!)\}$

- $\theta = \log \lambda$, $\lambda = e^\theta$, $b(\theta) = e^\theta$, $\phi = 1$, $V(\mu) = \mu$, Link = Log
- Log-lik: $\ell(\lambda) = \sum y_i \log \lambda - n\lambda - \sum \log(y_i!)$
- Score: $U(\lambda) = \frac{n(\bar{y} - \lambda)}{\lambda}$, Fisher: $\mathcal{I}(\lambda) = \frac{n}{\lambda}$

Exponential Distribution $Exp(\lambda)$

Rate parameterization.

- PDF: $f(y) = \lambda e^{-\lambda y}$, $y > 0$
- Mean: $E(Y) = 1/\lambda = \mu$, Variance: $\text{Var}(Y) = \mu^2$
- CDF: $F(y) = 1 - e^{-\lambda y}$
- MGF: $M(t) = \frac{\lambda}{\lambda - t}$, $t < \lambda$

Exp Form (mean $\mu = 1/\lambda$): $f(y|\mu) = \exp \left\{ -\frac{y}{\mu} - \log \mu \right\}$

- $\theta = -1/\mu = -\lambda$, $b(\theta) = -\log(-\theta) = \log \mu$
- $\phi = 1$, $V(\mu) = \mu^2$, Link = $-1/\mu$
- Log-lik: $\ell(\lambda) = n \log \lambda - \lambda \sum y_i$
- Score: $U(\lambda) = \frac{n}{\lambda} - \sum y_i$, Fisher: $\mathcal{I}(\lambda) = \frac{n}{\lambda^2}$

Gamma Distribution $\text{Gamma}(\alpha, \beta)$

Shape α , rate β . Mean $\mu = \alpha/\beta$.

- PDF: $f(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}$, $y > 0$
- Mean: $E(Y) = \alpha/\beta = \mu$, Variance: $\text{Var}(Y) = \alpha/\beta^2$
- MGF: $M(t) = \left(\frac{\beta}{\beta-t}\right)^\alpha$, $t < \beta$

Exp Form:

- $\theta = -\beta = -\alpha/\mu$, $b(\theta) = -\log(-\theta)$
- $\phi = 1/\alpha$, $V(\mu) = \mu^2$, Link = $-1/\mu$
- Note: $\text{Var}(Y) = \phi V(\mu) = \frac{\mu^2}{\alpha}$
- Log-lik (α fixed): $\ell(\beta) = n\alpha \log \beta - \beta \sum y_i + \text{const}$
- Score: $U(\beta) = \frac{n\alpha}{\beta} - \sum y_i$, Fisher: $\mathcal{I}(\beta) = \frac{n\alpha}{\beta^2}$

Beta Distribution $\text{Beta}(\alpha, \beta)$

Two-parameter EF (not standard 1-parameter EDM form unless one parameter is fixed).

- PDF: $f(y) = \frac{1}{B(\alpha, \beta)} y^{\alpha-1} (1-y)^{\beta-1}$, $y \in (0, 1)$
- $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
- Mean: $E(Y) = \frac{\alpha}{\alpha+\beta}$, Variance: $\text{Var}(Y) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
- Mode: $\frac{\alpha-1}{\alpha+\beta-2}$ (for $\alpha, \beta > 1$)
- Special cases: $\text{Beta}(1, 1) = U(0, 1)$; $\text{Beta}(1/2, 1/2) = \text{arc-sine}$

Exp Form (2-parameter EF):

$$f(y|\alpha, \beta) = \exp\{(\alpha-1)\log y + (\beta-1)\log(1-y) - \log B(\alpha, \beta)\}$$

- Natural params: $\eta_1 = \alpha - 1$, $\eta_2 = \beta - 1$
- Sufficient stats: $T_1(y) = \log y$, $T_2(y) = \log(1-y)$
- $A(\alpha, \beta) = \log B(\alpha, \beta) = \log \Gamma(\alpha) + \log \Gamma(\beta) - \log \Gamma(\alpha + \beta)$
- $E[\log Y] = \psi(\alpha) - \psi(\alpha + \beta)$, $E[\log(1-Y)] = \psi(\beta) - \psi(\alpha + \beta)$
(via $\partial A / \partial \alpha$, $\partial A / \partial \beta$)

- $h(y) = 1$ on $(0, 1)$ (support free of params $\Rightarrow$ regular EF)

Conjugacy: Beta is the conjugate prior for Bernoulli/Binomial p :

- Prior $p \sim \text{Beta}(\alpha, \beta)$, likelihood $X \sim \text{Bin}(n, p)$
- Posterior $p|X \sim \text{Beta}(\alpha + x, \beta + n - x)$
- Posterior mean (Bayes estimator under squared loss): $\frac{\alpha+x}{\alpha+\beta+n}$

Construction: If $X \sim \text{Gamma}(\alpha, \theta)$, $Y \sim \text{Gamma}(\beta, \theta)$ independent, then $\frac{X}{X+Y} \sim \text{Beta}(\alpha, \beta)$.

Order stats: If $U_1, \dots, U_n \stackrel{iid}{\sim} U(0, 1)$, then $U_{(k)} \sim \text{Beta}(k, n - k + 1)$.

Chi-square Distribution $\chi^2(k)$

Special case of Gamma with $\alpha = k/2, \beta = 1/2$.

- PDF: $f(y) = \frac{1}{2^{k/2}\Gamma(k/2)} y^{k/2-1} e^{-y/2}$, $y > 0$
- Mean: $E(Y) = k$, Variance: $\text{Var}(Y) = 2k$
- MGF: $M(t) = (1 - 2t)^{-k/2}$, $t < 1/2$

Exp Form (fixed k):

- $\theta = -1/2$ (fixed), $\phi = 2/k$
- $b(\theta) = -\log(-2\theta)$ (since $\mu = k = -1/\theta$)
- $\text{Var}(Y) = \phi \mu^2 = (2/k)k^2 = 2k$
- Log-lik: $\ell(k) = \frac{k-2}{2} \sum \log y_i - \frac{nk}{2} \log 2 - n \log \Gamma(\frac{k}{2}) + \text{const}$
- Score: $U(k) = \frac{1}{2} \sum \log y_i - \frac{n}{2} \log 2 - \frac{n}{2} \psi(\frac{k}{2})$
- Fisher: $\mathcal{I}(k) = \frac{n}{4} \psi'(\frac{k}{2})$

Key Formulas

Score (single obs): $U(\theta) = \frac{y-b'(\theta)}{\phi} = \frac{y-\mu}{\phi}$

Fisher Info (single obs): $\mathcal{I}(\theta) = \frac{b''(\theta)}{\phi} = \frac{V(\mu)}{\phi}$